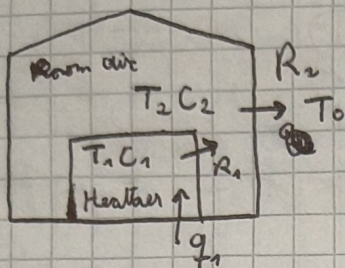


Fallen #2

①



$$C_1 \frac{dT_1}{dt} = q_0 - \frac{T_1 - T_2}{R_1}$$

$$C_2 \frac{dT_2}{dt} = \frac{T_1 - T_2}{R_1} - \frac{T_2 - T_0}{R_2}$$

$$X = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}, \quad u = q_0, \quad d = T_0$$

$$\dot{X} = \begin{bmatrix} \dot{T}_1 \\ \dot{T}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} & \frac{1}{R_1 C_1} \\ \frac{1}{R_1 C_2} & -\frac{1}{C_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{C_1} \\ 0 \end{bmatrix} q_0 + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{R_2 C_2} \end{bmatrix} d}_0$$

$$y = [0 \quad 1] \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = T_2$$

$$R_1 = 2^\circ\text{C/W}$$

$$R_2 = 5^\circ\text{C/W}$$

$$C_1 = 500 \text{ J/}^\circ\text{C}$$

$$C_2 = 800 \text{ J/}^\circ\text{C}$$

$$\frac{T_2}{q_0} = \frac{C_2 s + \left(\frac{1}{R_1} + \frac{1}{R_2} \right)}{(C_1 s + \frac{1}{R_1})(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}) - \left(\frac{1}{R_1} \right)^2} = \frac{800 s + 0,7}{400000 s^2 + 750 s + 0,1}$$

$$t_{sla} (s_{im}) = 40000 \text{ s} \rightarrow 100 \text{ m} \rightarrow T = 400 \text{ s}$$

$$\frac{0,002 s + 1,75 \times 10^{-6}}{s^2 + 0,001875 s + 0,25 \times 10^{-6}} \rightarrow \frac{0,67 z - 0,4735}{z^2 - 1,445 z + 0,4733}$$

$$ess = 0 \quad T = 400 \text{ s}$$

$$t_{sd} = 95\% = 38000 \text{ s} \quad \gamma = 1,2$$

$$P_1 = 0,9431; P_2 = 0,5018$$

$$C = 0,7067$$

$$-1,445 = -2 e^{-\gamma \omega_n T_m} \cos(\omega_n T \sqrt{1 - \gamma^2})$$

$$0,4733 = e^{-\gamma \omega_n T}$$

$$\omega_n = 782,05 \quad \gamma = 1 \times 10^{-5}$$

$$P_1 = \frac{1,445 - e^{-P_2 T}}{T}$$

$$P_2 = 1,998 \times 10^{-3}$$

$$P_1 = -48,75 \times 10^{-6}$$

$$\frac{0,67 z^{-1} - 0,4735 z^{-2}}{1 - 1,445 z^{-1} + 0,4733 z^{-2}}$$

$$z^{-2} - \frac{P_1 T}{z} + \frac{P_2 T}{z} = (e^{-P_1 T} \cdot e^{-P_2 T}) z - (e^{-P_1 T} \cdot e^{-P_2 T})$$

$$e^{-P_1 T} \cdot e^{-P_2 T} = 0,4733 \quad (e^{-P_1 T} + e^{-P_2 T}) = 1,445$$

$$\omega_n = \frac{\ln(1,445)}{-2 \gamma T_m}$$

$$-2 e^{-\gamma \omega_n T} \cos\left(\frac{\ln(1,445)}{-2 \gamma} \sqrt{1 - \gamma^2}\right) = 1,445$$

$$\frac{0,67z - 0,4735}{z^2 - 1,445z + 0,4733}$$

$$T = 400s \quad T_{ESS} = 0$$

$$t_{sd} = 5000s$$

$$z_d = 0,35$$

$$z = 0,9432$$

1I

$$W_{nd} = 4$$

$$0,00266$$

$$z = 0,5018$$

$$z_1 \rightarrow z_2$$

$$\frac{4}{5000 \cdot 0,3}$$

$$P_1 = -1,4532$$

$$P.D. = z^2 - 1,4532z + 0,5281 \rightarrow \text{Polo D.} = 0,7266 + 0,0123j$$

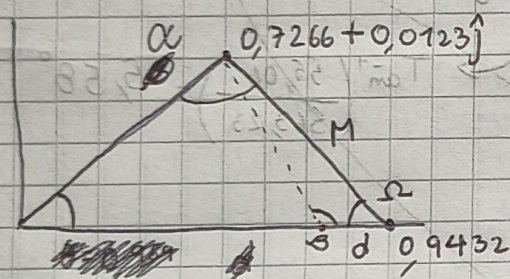
$$P_2 = 0,5281$$

$$\frac{0,67z - 0,4735}{(z^2 - 1,445z + 0,4733)(z - 1)} = 0,9705 + 0,665j \rightarrow \tan^{-1}\left(\frac{0,665}{0,9705}\right) = 34,42^\circ$$

Polo D.

$$\alpha = 180 - 34,42 = 145,58^\circ$$

$$\approx 3^\circ$$



$$M = \sqrt{(0,9432 - 0,7266)^2 + 0,0123^2} = 0,2169$$

$$\Omega = \tan^{-1}\left(\frac{0,0123}{(0,9432 - 0,7266)}\right) = 3,25^\circ$$

$$\beta = 145,58^\circ$$

$$\frac{\sin(0,9432)}{0,2169} = \frac{\sin(145,58^\circ)}{d}$$

$$d = 0,9432$$

$$180 - 48,6 - 3,25 = 128,15^\circ \quad \beta$$

$$d' = 0,2068$$

$$\frac{\sin(128,15^\circ)}{0,2169} = \frac{\sin(48,6^\circ)}{d}$$

$$0,9432 - 0,2068 = 0,7364$$

$$\frac{K(z - 0,9432)^3}{(z - 0,7364)^3}$$

$$\frac{0,67z - 0,4735}{(z^2 - 1,445z + 0,4733)(z - 1)} \cdot \frac{K(z - 0,9432)^3}{(z - 0,7364)^3} = K(-3076,13 + 51,183j)$$

Polo D.

$$K^2(3076,13^2 + 51,183^2) = 1$$

$$K = 3,25 \times 10^{-4}$$

$$\frac{3,25 \times 10^{-4} (z - 0,9432)^3}{(z - 0,7364)^3}$$

2a

$$ess = 0$$

$$T = 400 \text{ s}$$

$$G(z) = \frac{0,67 \bar{z}^{-1} - 0,4735 \bar{z}^2}{1 - 1,445 \bar{z}^{-1} + 0,4733 \bar{z}^{-2}}$$

$$\frac{0,67 \bar{z}^{-1} (1 + 0,4735 \bar{z}^{-1})}{(1 - 0,9431 \bar{z}^{-1})(1 - 0,5018 \bar{z}^{-1})}$$

$$P(z) = 1$$

$$q = 0$$

$$R(z) = \frac{P(z)}{(1 - \bar{z}^{-1})^{q+1}} = \frac{1}{(1 - \bar{z}^{-1})^1}; \quad U(z) = F(z) \cdot \frac{R(z)}{G(z)}$$

$$U(z) = b_0 + b_1 \bar{z}^{-1} = F(z) \frac{1}{(1 - \bar{z}^{-1})} \cdot \frac{(1 - 0,9431 \bar{z}^{-1})(1 - 0,5018 \bar{z}^{-1})}{0,67 \bar{z}^{-1} (1 + 0,4735 \bar{z}^{-1})}$$

$$F(z) = F_1 \bar{z}^{-1} + F_2 \bar{z}^{-2} \rightarrow 2 \bar{z}^{-1} - \bar{z}^{-2}$$

$$1 - F_1 \bar{z}^{-1} + F_2 \bar{z}^{-2} = (1 - \bar{z}^{-1}) N(z)$$

$$N(z) = 1 - \bar{z}^{-1}$$

$$F_1 = 2$$

$$F_2 = -1$$

$$G_D(z) = \frac{F(z)}{G(z) (1 - \bar{z}^{-1}) N(z)} = \frac{(2 \bar{z}^{-1} - \bar{z}^{-2}) (1 - 0,9431 \bar{z}^{-1}) (1 - 0,5018 \bar{z}^{-1})}{0,67 \bar{z}^{-1} (1 + 0,4735 \bar{z}^{-1}) (1 - \bar{z}^{-1}) (1 - \bar{z}^{-1})}$$

$$= \frac{\cancel{\bar{z}^{-1}} (2 - \bar{z}^{-1}) (1 - 0,9431 \bar{z}^{-1}) (1 - 0,5018 \bar{z}^{-1})}{\cancel{0,67 \bar{z}^{-1}} (1 + 0,4735 \bar{z}^{-1}) (1 - 2 \bar{z}^{-1} + \bar{z}^{-2})} \cdot \cancel{0,67}$$

$$= \frac{((2 - \bar{z}^{-1}) (1 - 0,9431 \bar{z}^{-1}) (1 - 0,5018 \bar{z}^{-1})) / 0,67}{(1 + 0,4735 \bar{z}^{-1}) (1 - 2 \bar{z}^{-1} + \bar{z}^{-2})}$$

$$\frac{(2 - 2,8862 \bar{z}^{-1} + 0,9431 \bar{z}^{-2}) (1 - 0,5018 \bar{z}^{-1})}{(1 + 0,4735 \bar{z}^{-1}) (1 - 2 \bar{z}^{-1} + \bar{z}^{-2})} \cdot \cancel{0,67}$$

4

$$T = 0,05s$$

$$t_{sd} = 0,5s$$

$$p_1 = -1,340$$

$$p_2 = 0,449$$

$$\gamma_d = 0,3$$

$$W_{hd} = \frac{4}{0,3 \cdot 0,5} = 26,666$$

$$z^2 - 1,340z + 0,449 = P.D.$$

$$\rightarrow 0,67 + 0,01j \text{ P.D.}$$

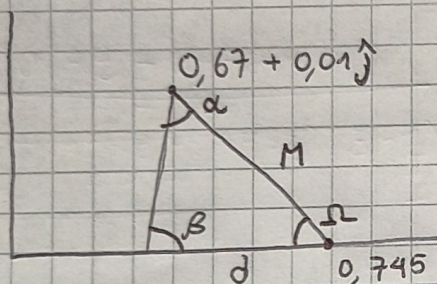
$$\frac{0,255}{(z - 0,745)(z - 1)^2} \Big|_{\text{Polo D.}} = -30,34 - 5,934j$$

↓

$$z = 0,745$$

$$\rightarrow 11,06^\circ \rightarrow \alpha = 180^\circ - 11,06^\circ = 168,93^\circ$$

$$168,93 \begin{array}{l} \text{60} \\ \approx 3, \\ \div 3 = 56,311^\circ, \alpha \end{array}$$



$$M = \sqrt{(0,745 - 0,67)^2 + 0,01^2} = 0,0757$$

$$\Omega = \tan^{-1} \left(\frac{0,01}{(0,745 - 0,67)} \right) = 7,595^\circ$$

$$\beta = 180^\circ - 7,595^\circ - 56,311^\circ = 116,094^\circ$$

$$\frac{\sin(116,094^\circ)}{0,0757} = \frac{\sin(56,311^\circ)}{d}$$

$$d = 0,0701$$

$$0,745 - 0,0701 = 0,6749$$

$$K = \frac{(z - 0,745)^3}{(z - 0,6749)^3}$$

$$\frac{0,255}{(z - 0,745)(z - 1)^2} \cdot \frac{K(z - 0,745)^3}{(z - 0,6749)^3} \Big|_{\text{Polo D.}} = 9698,48 - 5,491j$$

$$K^2(9698,48^2 + 5,49^2) = 1$$

$$K = 1,031 \times 10^{-4}$$

$$\boxed{1,031 \times 10^{-4} \frac{(z - 0,745)^3}{(z - 0,6749)^3}}$$

0.5

3a

$$\frac{0,255 z^{-1}}{1 - 0,745 z^{-1}}$$

$$T = 0,03s$$

$$e_{ss} = 0$$

$$P(z) = T z^{-1}$$

$$q = 1$$

$$R(z) = \frac{P(z)}{(1 - z^{-1})^2} = \frac{0,03 z^{-1}}{(1 - z^{-1})^2}$$

$$U(z) = \frac{F(z)}{G(z)} R(z); U(z) = b_0 + b_1 z^{-1}$$

$$F(z) = \frac{0,03 z^{-1}}{(1 - z^{-1})^2} \cdot \frac{(1 - 0,745 z^{-1})}{0,255 z^{-1}}$$

$$F(z) = F_1 z^{-1} + F_2 z^{-2}$$

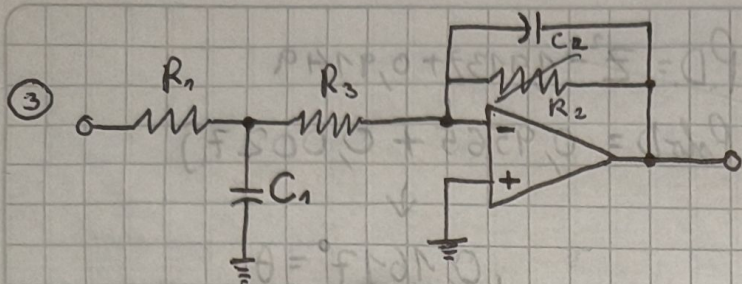
$$N(z) = 1$$

$$1 - F_1 z^{-1} - F_2 z^{-2} = (1 - 2z^{-1} + z^{-2}) N(z)$$

$$F_1 = 2$$

$$F_2 = -1$$

$$G_0(z) = \frac{2z^{-1} - z^{-2}(1 - 0,745z^{-1})}{0,255z^{-1}(1 - 2z^{-1} + z^{-2})} = \frac{z^{-1}(2 - z^{-1})(1 - 0,745z^{-1})}{0,255z^{-1}(1 - 2z^{-1} + z^{-2})}$$



$$X_1 = V_{ij}(C_2)$$

$$X_2 = V_z \text{ (out)}$$

$$U = V_i$$

$$iR_2 = e^2$$

$$\frac{U - X_1}{R_1} = C_1 \dot{X}_1 + \frac{X_1}{R_3}$$

$$\dot{X}_1 = \frac{1}{C_1} \left(\frac{U - X_1}{R_1} - \frac{X_1}{R_3} \right)$$

$$\frac{X_1}{R_3} + C_2 \dot{X}_2 + iR_2 = 0$$

$$iR_2 = X_1^2 \quad C_2 \dot{X}_2 = -\frac{X_1}{R_3} - X_1^2$$

$$\dot{X}_1 = \frac{1}{C_1} \left(\frac{U - X_1}{R_1} - \frac{X_1}{R_3} \right)$$

$$X_2 = X_{20} + \delta X_2$$

$$\dot{X}_2 = -\frac{1}{R_3 C_2} X_1 - \frac{1}{C_2} X_1^2$$

$$X_2^2 = X_{20}^2 + 2 X_{20} \delta X_2$$

$$\dot{X}_2 = \frac{1}{R_3 C_2} X_1 - \frac{2 X_{20}}{C_2} X_1$$

$$y = X_2$$

$$\frac{1}{R_1 R_3 C_1 C_2} s^2 + \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_1} + \frac{2 X_{20}}{C_2} \right) s + \frac{2 X_{20}}{R_1 R_3 C_1 C_2} = \frac{-20}{s^2 + 20003s + 60000}$$

$$R_1 = 10 \text{ K}\Omega; R_3 = 5 \text{ K}\Omega$$

$$C_1 = 100 \mu\text{F}; C_2 = 10 \mu\text{F}$$

$$t_{sla} = 2s \rightarrow 10\text{ms} \Rightarrow T = 0,2$$

$$\frac{-1,93 \times 10^5}{2 - 0,942} = \frac{-1,93 \times 10^5 z^{-1}}{1 - 0,942 z^{-1}}$$

6

$$\frac{-1,935 \times 10^{-5}}{(z-0,942)(z-1)^3}$$

$$\downarrow$$

$$z = 0,942$$

$$t_{sd} = 1,2s$$

$$T = 0,02s$$

$$z_d = 0,7$$

$$W_{nd} = \frac{4}{12 \cdot 0,7} = 4,761$$

$$P_1 = -1,871$$

$$P_2 = 0,8752$$

$$P.D. = z^2 - 1,871z + 0,8752$$

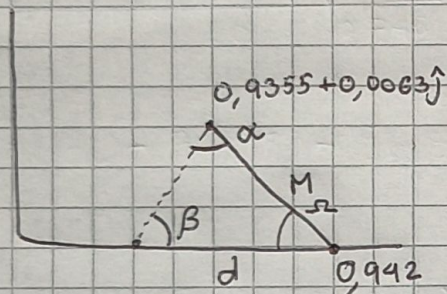
$$z = 0,9355 + 0,0063j \text{ Pol.D.}$$

$$\left. \frac{-1,935 \times 10^{-5}}{(z-0,942)(z-1)^3} \right|_{\text{Pol.D.}} = 0,29 + 0,418j$$

$$\searrow 55,248^\circ$$

$$\alpha = 180^\circ - 55,248^\circ = 124,75$$

$$\downarrow \div 3 = 41,6^\circ, \alpha$$



$$M = \sqrt{(0,942 - 0,9355)^2 + (0,0063)^2} = 0,0090$$

$$\omega = \tan^{-1} \left(\frac{0,0063}{(0,942 - 0,9355)} \right) = 44,104^\circ$$

$$\beta = 180^\circ - 44,104^\circ - 41,6^\circ = 94,296^\circ$$

$$\frac{\sin(94,26^\circ)}{0,0090} = \frac{\sin(41,6^\circ)}{d}$$

$$d = 5,9918 \times 10^{-3}$$

$$0,942 - 5,9918 \times 10^{-3} = 0,936$$

$$\frac{K(z-0,942)^3}{(z-0,936)^3}$$

$$\frac{-1,935 \times 10^{-5}}{(z-0,942)(z-1)^3} \cdot \frac{K(z-0,942)^3}{(z-0,936)^3} = K(22,993 + 1,976j)$$

$$K^2(22,993^2 + 1,976^2) = 0,0433$$

$$\boxed{0,0433 \frac{(z-0,942)^3}{(z-0,936)^3}}$$

58

6a

$$\frac{-1,93 \times 10^{-5} z^{-1}}{1 - 0,942 z^{-1}}$$

$$T = 0,02 \text{ s}$$

$$ess = 0$$

$$P(z) = \frac{1}{2} T^2 z^{-1} (1 + z^{-1})$$

$$q = 2$$

$$\frac{G_0(z) G(z)}{1 + G_0(z) G(z)} = z^{-1}$$

$$\frac{z^{-1}}{1 + G_0(z) G(z)} = \frac{z^{-1}}{1 - 0,942 z^{-1} - 1,93 \times 10^{-5} z^{-1}}$$

$$G_0(z) G(z) = \frac{z^{-1}}{1 - z^{-1}}$$

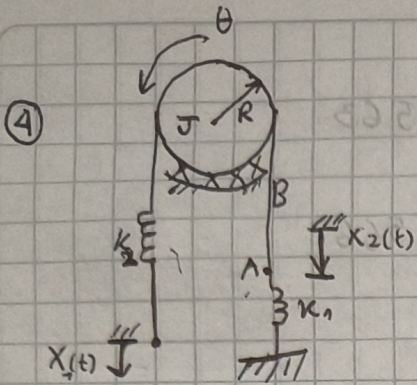
$$G_0(z) = \frac{z^{-1}}{1 - z^{-1}} \cdot \frac{1 - 0,942 z^{-1}}{-1,93 \times 10^{-5} z^{-1}}$$

$$G_0(z) = \frac{z^{-1} - 0,942 z^{-2}}{-1,93 \times 10^{-5} z^{-1} + 1,93 \times 10^{-5} z^{-2}}$$

$$0,84 = -X \cdot 0,02 = -0,02X \quad 0,84 = -X \cdot 0,02 = -0,02X$$

$$\frac{z^{-1} - 0,942 z^{-2}}{z^{-2} (1,93 \times 10^{-5} - 1,93 \times 10^{-5} z^{-1})} = \frac{z^{-1} - 0,942 z^{-2}}{z^{-2} (1,93 \times 10^{-5} (1 - z^{-1}))}$$

7



$$x_1 = R\theta; \quad x_2 = R\theta$$

$$J\ddot{\theta} = -B\dot{\theta} - \tau_1 - \tau_2$$

$$\tau_1 = K_1(R\theta)R = K_1R^2\theta$$

$$\tau_2 = K_2(R\theta)R = K_2R^2\theta$$

$$J\ddot{\theta} + B\dot{\theta} + (K_1 + K_2)R^2\theta = 0$$

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{(K_1 + K_2)R^2}{J} & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y = \begin{bmatrix} R & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$R = 0,1 \text{ m}; \quad J = 0,2 \text{ Kg}\cdot\text{m}^2; \quad B = 0,1 \text{ Nms}$$

$$K_1 = 50 \text{ N/m}; \quad K_2 = 30 \text{ N/m}$$

$$\frac{X(s)}{T(s)} = \frac{R}{Js^2 + Bs + (K_1 + K_2)R^2}$$

$$t_{sla} = 20 \text{ s}$$

$$T = \frac{0,2}{0,25}$$

$$\frac{0,1}{s^2 + 0,5s + 4} \leftarrow \frac{0,1}{0,2s^2 + 0,15s + 0,8} \quad \begin{matrix} /0,2 \\ /0,2 \end{matrix}$$

$$\frac{0,00937 \bar{z}^1 + 0,00937 \bar{z}^2}{1 - 1,754 \bar{z}^1 + 0,904 \bar{z}^2}$$

$$t_{sd} = 18 \text{ s}$$

$$ess = 0$$

$$z_d = 0,75$$

$$P_1 = -1,913$$

$$W_{nd} = 0,2963$$

$$P_2 = 0,9149$$

$$P.D. = z^2 - 1,913z + 0,9149$$

↓

$$\text{Polo D.} = 0,9565 + 0,0027j$$

↓

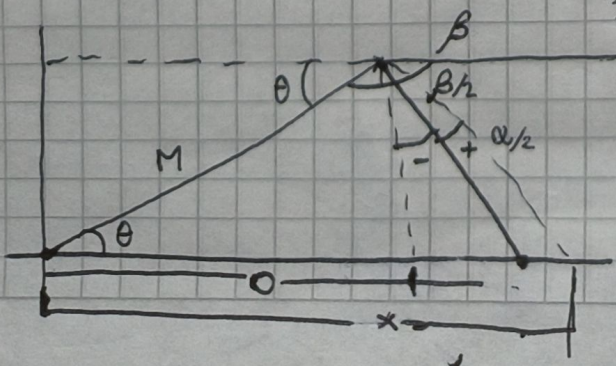
$$0,1617^\circ \theta$$

$$T = 0,2 \text{ s}$$

$$-1 = 0$$

$$\frac{0,00937 \bar{z} + 0,00937}{\bar{z}^2 - 1,754 \bar{z} + 0,904 (1 - \bar{z})^2} = 67,84 + 8,340j$$

$$0,9565 + 0,0027j \quad \downarrow \quad 7^\circ = \alpha$$



$$P_1 = 0,9565$$

$$P_2 = 0,9149$$

$$+ = 93,919$$

$$B = 180 - 8,1617$$

$$B = 171,83$$

$$- = 86,915$$

$$B_2 = 89,915$$

8

$$\alpha = 180^\circ - 7^\circ = 173^\circ \quad \left| \begin{array}{l} 60 \\ \approx 3 \end{array} \right.$$

$$\downarrow \Rightarrow$$

$$\alpha = 157,7^\circ$$

$$\beta = 180^\circ - \theta = 179,83^\circ$$

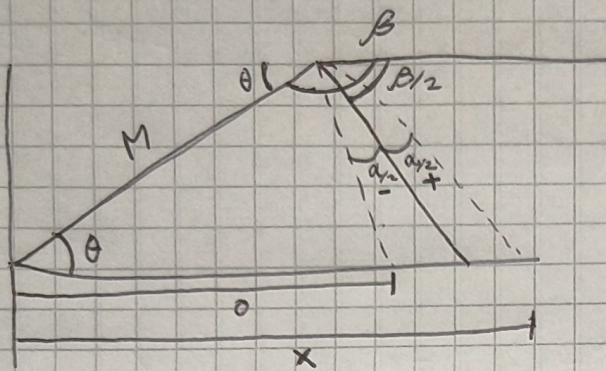
$$\beta/2 = 89,915^\circ$$

$$\alpha/2 = 28,85^\circ$$

$$\beta/2 + \alpha/2 = 118,765^\circ$$

$$\beta/2 - \alpha/2 = 61,065^\circ$$

$$M = 0,9565$$



$$\frac{\text{Sen}(118,765^\circ)}{0,9565} = \frac{\text{Sen}(61,065^\circ)}{o}$$

$$o = 0,9549$$

$$x = 0,9581$$

$$\frac{\text{Sen}(61,065^\circ)}{0,9565} = \frac{\text{Sen}(118,765^\circ)}{x}$$

$$\frac{K(z - 0,9549)^3}{(z - 0,9581)^3} \cdot \frac{(0,00937z + 0,00937)}{(z^2 - 1,754z + 0,904)(z-1)^3} \Bigg|_{\text{P\&D}} = -68 - 3,68j$$

$$K^2(68^2 + 3,68^2) = 1$$

$$K = 0,01468$$

$$\boxed{0,01468 \frac{(z - 0,9549)^3}{(z - 0,9581)^3}} \quad \triangleleft$$

8a

$$\frac{0,00937z^{-1} + 0,0093z^{-2}}{1 - 1,754z^{-1} + 0,904z^{-2}}$$

$$T = 0,2s$$

$$ess = 0$$

$$P(z) = 1$$

$$q = 0$$

$$R(z) = \frac{1}{(1 - z^{-1})}$$

$$\frac{0,00937z^{-1}(1 - z^{-1})}{1 - 1,754z^{-1} + 0,904z^{-2}}$$

$$U(z) = \frac{R(z)}{G(z)} = \frac{1 - 1,754z^{-1} + 0,904z^{-2}}{0,00937z^{-1}(1 - z^{-1})(1 - z^{-1})}$$

$$U(z) = b_0 + b_1z^{-1}$$

$$F(z) = F_1z^{-1} + F_2z^{-2}$$

$$1 - F(z) = 1 - F_1z^{-1} - F_2z^{-2} = (1 - z^{-1})$$

$$\alpha_1 = -1,754 \quad B_0 = 0,0093 \quad B_2 = 0$$

$$\alpha_2 = 0,904 \quad B_1 = 0,0093$$

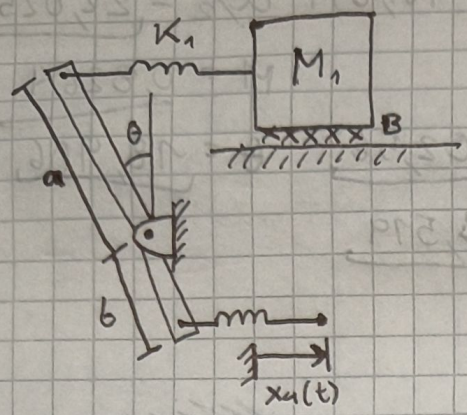
$$q_0 = \frac{1}{0,0093 + 0,0093} = 53,763$$

$$q_1 = q_0 \cdot \alpha_1 = -94,3 \quad ; \quad q_2 = q_0 \cdot \alpha_2 = 48,60$$

$$p_0 = q_0 B_0 = 0,499 \quad ; \quad p_1 = q_0 B_1 = 0,499 \quad ; \quad p_2 = 0$$

$$G_D(z) = \frac{53,763 + (-94,3)z^{-1} + 48,6z^{-2}}{1 - (0,499z^{-1} + 0,499z^{-2})}$$

5



$$x_1 = a\theta, \quad x_2 = b\theta$$

$$F_1 = K_1(x - a\theta); \quad F_2 = K_2(b\theta)$$

$$M_1 \ddot{x} + B \dot{x} + K_1(x - a\theta) = f_a(t)$$

$$K_1(x - a\theta)a - K_2(b\theta)b = 0$$

$$K_1 a x + K_1 a^2 \theta - K_2 b^2 \theta = 0$$

$$\theta = \frac{K_1 a}{K_1 a^2 + K_2 b^2} x \quad \left| \quad M_1 \ddot{x} + B \dot{x} + K_1 x - K_1 a \left(\frac{K_1 a}{K_1 a^2 + K_2 b^2} x \right) = f_a \right.$$

$$K_{eq} = K_1 - \frac{K_1^2 a^2}{K_1 a^2 + K_2 b^2} \rightarrow M_1 \ddot{x} + B \dot{x} + K_{eq} x = f_a$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{K_{eq}}{M_1} & -\frac{B}{M_1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M_1} \end{bmatrix} f_a$$

$K_1 = 100 \text{ N/m}$
 $K_2 = 150 \text{ N/m}$
 $a = 0,2 \text{ m}$
 $b = 0,3 \text{ m}$
 $M_1 = 2 \text{ Kg}$
 $B = 5 \frac{\text{N}\cdot\text{s}}{\text{m}}$

$$y = \begin{bmatrix} \frac{K_1 a}{K_1 a^2 + K_2 b^2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{\Theta(s)}{F_a(s)} = \frac{\frac{K_1 a}{K_1 a^2 + K_2 b^2}}{M_1 s^2 + B s + K_{eq}} = \frac{10}{s^2 + 2,5s + 38,57}$$

$$t_{sla} = 4,5 \text{ s}$$

$$T = 0,2 \text{ s}$$

$$\frac{0,1157 z^{-1} + 0,0989 z^{-2}}{1 - 0,544 z^{-1} + 0,6065 z^{-2}}$$

$$t_{sd} = 4,2 \text{ s}$$

$$P_1 = -1,5639$$

$$P.D. = z^2 - 1,5639z + 0,6832$$

$$z_d = 0,5$$

$$P_2 = 0,6832$$

$$\text{Rolo.D.} = 0,7819 + 0,2678j$$

$$W_{nd} = 1,9047$$

$$e_{ss} = 0$$

$$18,906^\circ \quad M = 0,8264$$

$$T = 0,2 \text{ s}$$

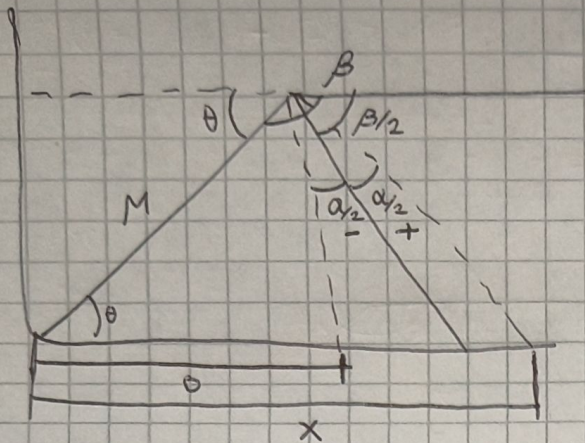
$$\frac{0,1157 z + 0,0989}{(z^2 - 0,544 z + 0,6065)(z - 1)^2} = 0,0667 + 1,991j$$

$$0,7819 + 0,2678j$$

$$88,08 \approx 2c$$

$$180 - 88,08 = 91,92 \text{ (60)}$$

$$\frac{1}{2} \rightarrow 46^\circ$$



$$\beta = 180 - \theta = 161,094^\circ$$

$$\alpha/2 = 23^\circ$$

$$\beta/2 = 80,54^\circ$$

$$M = 0,8264$$

$$\beta/2 + \alpha/2 = 103,54^\circ$$

$$\theta = 18,906^\circ$$

$$\beta/2 - \alpha/2 = 57,54^\circ$$

$$\frac{\text{Sen}(103,54^\circ)}{0,8264} = \frac{\text{Sen}(57,54^\circ)}{0}$$

$$0 = 0,717$$

$$\frac{\text{Sen}(57,54^\circ)}{0,8264} = \frac{\text{Sen}(103,54^\circ)}{x}$$

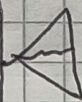
$$x = 0,952$$

$$\left. \frac{K(z - 0,717)^2}{(z - 0,952)^2} \cdot \frac{0,1157z - 0,0989}{(z^2 - 0,544z + 0,6065)(z - 1)^2} \right|_{\text{Polo } 0} = -0,050 + 0,262j$$

$$K^2(0,65^2 + 0,262^2) = 1$$

$$K = 3,75$$

$$3,75 \frac{(z - 0,717)^2}{(z - 0,952)^2}$$



$$\frac{0,1157 \bar{z}^{-1} + 0,0989 \bar{z}^{-2}}{1 - 0,544 \bar{z}^{-1} + 0,6065 \bar{z}^{-2}}$$

$$ess = 0$$

$$T = 0,2s$$

$$\frac{0,1157 \bar{z}^{-1} \cdot (1 + 0,8548 \bar{z}^{-1})}{1 - 0,544 \bar{z}^{-1} + 0,6065 \bar{z}^{-2}}$$

$$U(z) = b_0 + b_1 \bar{z}^{-1} + b_2 \bar{z}^{-2}$$

$$P(z) = \cancel{1} T \bar{z}^{-1}$$

$$q = 1$$

$$R(z) = \frac{0,2 \bar{z}^{-1}}{(1 - \bar{z}^{-1})^2}$$

$$U(z) = F(z) \frac{0,2 \bar{z}^{-1}}{(1 - \bar{z}^{-1})^2} \cdot \frac{(1 - 0,544 \bar{z}^{-1} + 0,6065 \bar{z}^{-2})}{0,1157 \bar{z}^{-1} (1 + 0,8548 \bar{z}^{-1})}$$

$$U(z) = b_0 + b_1 \bar{z}^{-1}$$

$$F(z) = F_1 \bar{z}^{-1} + F_2 \bar{z}^{-2}$$

$$N(z) = 1$$

$$F_1 = 2$$

$$F_2 = -1$$

$$1 - F_1 \bar{z}^{-1} - F_2 \bar{z}^{-2} = (1 - \bar{z}^{-1})^2 N(z) = 1 - 2 \bar{z}^{-1} + \bar{z}^{-2}$$

$$F(z) = 2 \bar{z}^{-1} - \bar{z}^{-2}$$

$$G_D(z) = \frac{2 \bar{z}^{-1} - \bar{z}^{-2}}{1 - 2 \bar{z}^{-1} + \bar{z}^{-2}} \cdot \frac{(1 - 0,544 \bar{z}^{-1} + 0,6065 \bar{z}^{-2})}{0,1157 \bar{z}^{-1} (1 + 0,8548 \bar{z}^{-1})}$$

$$= \frac{\bar{z}^{-1} [(2 - \bar{z}^{-1}) (1 - 0,544 \bar{z}^{-1} + 0,6065 \bar{z}^{-2})]}{(1 - \bar{z}^{-1})^2 \cancel{0,1157 \bar{z}^{-1}} (1 + 0,8548 \bar{z}^{-1})} \cdot \frac{0,1157}{\cancel{0,1157}}$$